

NEHA VALUTHUNDIL SHAJI

BSC (HONS) CREATIVE COMPUTING STUDENT | MACHINE LEARNING & DATA SCIENCE ENTHUSIAST

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PROFESSIONAL OBJECTIVE

Undergraduate student with strong interest in artificial intelligence research and demonstrated experience in machine learning experimentation, computer vision, and quantitative study design. Proven ability to investigate model performance through comparative evaluation and data-driven analysis. Seeking participation in MBZUAI UGRIP to develop rigorous AI research skills and contribute to collaborative research projects.

EDUCATION

BSc (Hons) Creative Computing, Bath Spa University, Ras Al Khaimah

(2023 - 2026)

CGPA: 3.3 / 4.0 (Years 1-2 completed)

Machine Learning (Year 3): 70% (First-Class)

Relevant Coursework: Machine Learning, Research Project

Award: Academic Excellence Award (Year 2, Semester 2)

TECHNICAL SKILLS

- **Programming:** Python, SQL, JavaScript.
- **ML & Data Tools:** NumPy, Pandas, Scikit-learn, TensorFlow, Keras, Jupyter Notebook, Git, Gradio.
- **Methods:** Regression, Classification, CNN, Transfer Learning, Data Preprocessing, Cross-Validation, Model Evaluation, Statistical Analysis, Experimental Design.

ACADEMIC & RESEARCH PROJECTS

- **Weather Image Classification (CNN & Transfer Learning)**

Investigated transfer learning effectiveness for small-scale weather image classification using MobileNetV2 with augmentation strategies and performance evaluation; deployed interactive inference interface.

Kaggle Link : <https://www.kaggle.com/code/neha005/weather-image-classification-ml-project>

- **Online Purchase Prediction (Classification)**

Conducted exploratory data analysis, class imbalance mitigation, and comparative benchmarking across Logistic Regression, KNN, Decision Tree, and Random Forest models.

Kaggle Link : <https://www.kaggle.com/code/neha005/02-classification-task>

- **Sports Car Price Prediction (Regression)**

Performed dataset cleaning and feature preparation (~1k records) and evaluated regression models using cross-validation to analyze predictive performance.

Kaggle Link : <https://www.kaggle.com/code/neha005/01-regression-task>

- **Immersive 360° Educational Storytelling (Research Project)**

Designed and conducted survey-based study (n=100) examining engagement and learning outcomes using statistical analysis and data interpretation.

Link:- https://docs.google.com/document/d/1BE_JSxqosQD70vdapbvH0fLbgYH2i325/edit?pli=1

RELEVANT CERTIFICATIONS

- **IBM:** Python for Data Science
- **Kaggle:** Intro & Intermediate Machine Learning; SQL for Data Science
- **LinkedIn Learning:** Probability Foundations for Data Science

RESEARCH INTERESTS

- Machine Learning, Computer Vision, Data Science, and Human-AI Interaction with emphasis on applied AI experimentation and real-world data driven research.